MILITARY SPECIFICATION

FIELD CHANGE KIT
FOR VARIABLE OUTPUT POWER CAPABILITY
FOR TRANSMITTER AN/URT-23(C)

This amendment forms a part of Military Specification MIL-F-28874, dated 27 May 1983, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

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3.7.2.1, last sentence: Delete and substitute: "In the data mode, the audio input shall be the NET TEST signal at zero decibels referred to one milliwatt (dBm) average power and 8.6 dBm PEP in each sideband."

3.7.2.2: Delete the text and substitute: "Power output controls located on the AM-3924( )/URT front panel shall permit the continuous adjustment of output power to desired levels between 10 mW PEP, or average, and the full output levels of TABLE I. Power control shall be provided in five switch-selected ranges of 10 dB each (50 dB total), with analog control providing adequate resolution within each range and sufficient overlap to ensure continuous control. Power output in the DATA modes in the 100 W to 1000 W range shall be fixed at the maximum level specified in TABLE I for DATA modes.".

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TABLE I, title: Delete and substitute: "Maximum power output". Delete last two entries in their entirety and substitute the following:

| | | |
|---|---|---|
| "F9 (TADIL A)[2/] double side band | 1000 W PEP<br>200 W average | 0, +2 dB<br>± 1 dB |
| F9j (TADIL A)[2/] SSB | 1000 W PEP<br>100 W average | 0, +2 dB<br>± 1 dB" |

TABLE I, footnote "[3/]": Delete in its entirety.

3.7.4b, last sentence: Delete the period and add: "for power control ranges below 100 W.".

3.7.5, line 5: Delete "toggle".

3.7.5.1: Delete the text and substitute: "Forward power delivery to a 50-ohm load shall be within ±1.5 dB of the indicated value when measured at the 1000 W, 100 W, 10 W, 1.0 W, and 0.1 W points on the appropriate meter range providing upscale indication. Maximum total meter error shall not exceed 3 dB at a forward power indication of 50 mW, or 6 dB at a forward indication of 10 mW, into a 50 ohm load.".

After "3.7.5.2", add the following new paragraphs:

"3.7.5.3 Final amplifier screen circuit protection. Positive protection shall be provided against damage to the AM-3924 screen bias components due to plate-to-screen arc-overs. Protection circuitry shall reset automatically when the arc extinguishes. Blowing of primary power fuses during plate-to-screen arc-overs is permissible.

"3.7.5.4 RF output monitor sample. A sample of the radio transmitter RF output shall be provided to a receptacle (BNC) located on the AM-3924( ) front panel. The sample port source impedance shall be nominally 50 ohms, and the sampling ratio shall be -60 dB, ±8 dB, relative to the RF output delivered to a 50 ohm load connected to the antenna terminals.".

3.7.6, after the last sentence, add: "In addition, the transmitter with field change kit installed shall conform to the EMC requirements of MIL-T-28706.".

FSC 5820



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TABLE II: Add the following new entries:

| Examination or test | Requirement paragraph | Test: paragraph | First article inspection | Production inspection |
|---|---|---|---|---|
| "Screen circuit protection | 3.7.5.3 | 4.7.9 | X | |
| RF output monitor sample | 3.7.5.4 | 4.7.10. | X | " |

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After "4.7.8", add the following new paragraphs:

"4.7.9 Final amplifier screen circuit protection. Tests shall bse performed to verify compliance with 3.7.5.3. Plate-to-screen arc-overs shall be simulated in AM-3924( ) final and driver tube stages.

"4.7.10 RF output monitor sample. Amplitude of the RF output monitor sample shall be measured into a 50-ohm load connected to the AM-3924C front panel monitor port.. Measurement shall be made with a single-tone 100 W RF output at operating frequencies of 2 MHz, 4 MHz, 8 MHz, 16 MHz, and 29.9999 MHz.".

Preparing activity:
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